

SYDNEY TECHNICAL HIGH SCHOOL

YEAR 12

ASSESSMENT TASK 2

MARCH 2008

MATHEMATICS

Time Allowed: 70 minutes

Instructions:

- Write your name and class at the bottom of this page
- Attempt all questions
- Show all necessary working
- Marks may be deducted for careless or badly arranged work
- Approved calculators may be used
- At the end of the examination hand in both the question paper and your answers
- Marks indicated are a guide only and may be varied if necessary
- Standard integrals are attached and may be removed for your convenience.

Name: _____ Teacher: _____

Question 1	Question 2	Question 3	Question 4	Question 5	Total
/11	/10	/11	/11	/11	/54

QUESTION 1 (11 Marks)

- i) Find the number of terms in the arithmetic sequence 2

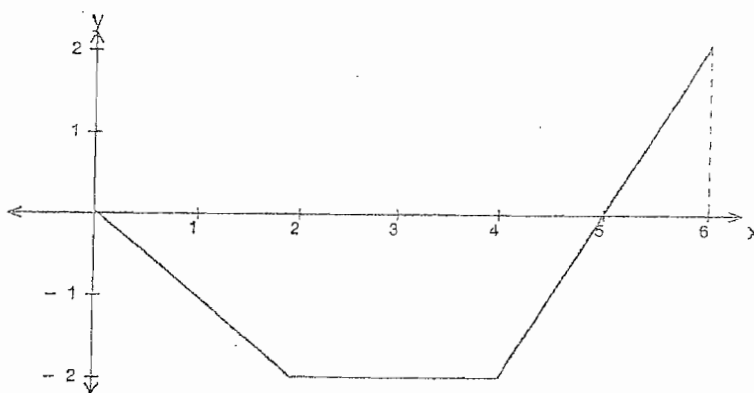
$$10, 6, 2, \dots, -102$$

- ii) Differentiate $y = \frac{3x^2}{x+5}$ and express your answer in simplest form 3

- iii) Find the primitive function of 2

$$2x^2 - \frac{1}{x^2}$$

- iv) The diagram represents a function $y = f(x)$. 2



Evaluate $\int_0^6 f(x) dx$.

- v) Find the equation of the curve $y = f(x)$ given that $f'(x) = 2x + 1$ and that the curve passes through $(1, 4)$ 2

QUESTION 2 (10 Marks) **Start a new page**

i) For a sequence it is given that

$$S_n = n^2 + 4n$$

a) Express S_{n-1} in terms of n **1**

b) Hence, or otherwise express T_n in terms of n **2**

c) Find the 10th term of the sequence **1**

ii) A person saved \$1000 the first year and \$200 more each subsequent year.
How many years will it take to save \$58000? **4**

iii) Evaluate $32 + 24 + 18 + \dots$ **2**

QUESTION 3 (11 Marks) **Start a new page**

A) Consider the function

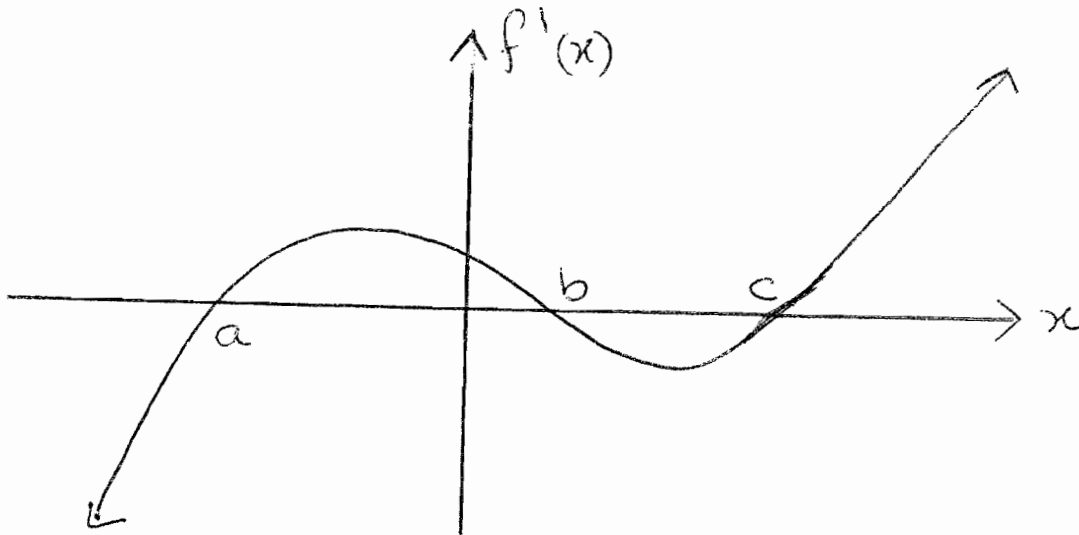
$$f(x) = x^3 + 9x^2 + 24x + 3$$

i) Find the co-ordinates of the stationary points of the curve $y = f(x)$ and determine their nature. 3

ii) Sketch the curve, clearly labelling any stationary points and the y – intercept 2

iii) For what vales of x is the curve decreasing? 1

B) This is a diagram of $y = f'(x)$



i) Write down the x values of any stationary points on $y = f(x)$ 1

ii) For what values of x is $y = f(x)$ increasing? 2

iii) Sketch a possible graph of $y = f(x)$ given that $y = f(x)$ passes through $(0, 2)$ 2

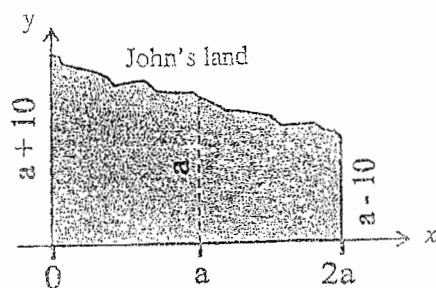
QUESTION 4 - (11 Marks) [Start a new page](#)

A) A couple borrow \$400,000 to purchase a house. They must repay the loan by equal quarterly instalments. Interest is charged at the rate of 8% p.a

- i) Write down the quarterly interest rate 1
- ii) Write an expression for A_1 , the amount owing after the first quarterly repayment. Let M be the amount repaid at the end of each quarter. 1
- iii) Show that the amount owing at the end of the first year is given by $400\,000(1.02)^4 - M(1 + 1.02 + 1.02^2 + 1.02^3)$ 2
- iv) Find the amount of each quarterly instalment if the loan is to be fully repaid in 12 years. (answer to the nearest dollar) 3

B)

The shaded area shown in the diagram below represents John's land. Its dimensions are given in terms of 'a'.



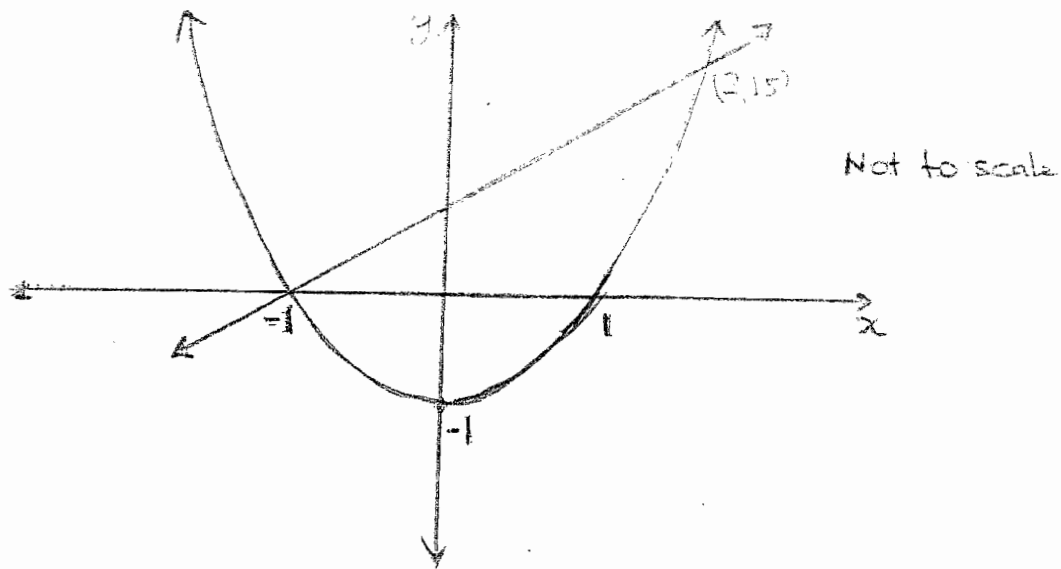
- i) Complete the table:

x	0	a	2a
y			

Given that the area of this land is 3200m^2 , use Simpson's rule with 3 function values to find an estimate for the value of 'a'

QUESTION 5 (11 Marks) Start a new page

A)



The diagram shows the curve $y = x^2 - 1$ and the line $y = 5x + 5$

- (i) Show that the line and curve intersect at the points $(-1, 0)$ and $(2, 15)$ 2
- (ii) Calculate the area between the curve and the line. 3

B) A piece of wire 24 cm long is cut into two pieces. Each is bent to form a square.

- i) If one piece is x cm long, write an expression for the length of the other piece 1
- ii) Show that the sum of the areas of the two squares is given by
- $$\left(\frac{x}{4}\right)^2 + \left(\frac{24-x}{4}\right)^2$$
- 2
- iii) Find the minimum area of the two squares 3

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Teacher's Name:

Student's Name/N^o:

Ex: 10.1

i) A.P. with $a = 10$ $d = -4$

$$T_n = a + (n-1)d$$

$$-102 = 10 + (-4)(n-1)$$

$$= 10 - 4n + 4$$

$$\therefore 4n = 116$$

$$n = 29$$

There are 29 terms

$$\text{ii) } y = \frac{3x^2}{x+5}$$

Quotient

$$\frac{u}{v}$$

where

$$u = 3x^2$$

$$v = x+5$$

$$u' = 6x$$

$$v' = 1$$

$$y' = \frac{u'v - v'u}{v^2}$$

$$= \frac{6x(x+5) - 1(3x^2)}{(x+5)^2}$$

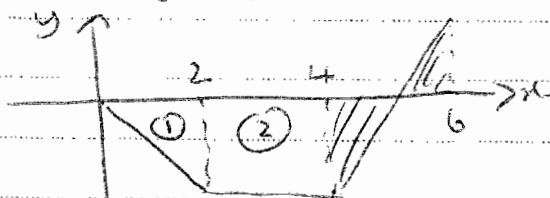
$$= \frac{6x^2 + 30x - 3x^2}{(x+5)^2}$$

$$= \frac{3x^2 + 30x}{(x+5)^2}$$

$$= \frac{3x(x+10)}{(x+5)^2}$$

(iii) Primitive of $2x^2 - \frac{1}{x^2}$

$$= \frac{2x^3}{3} + \frac{1}{x} + c$$

(iv) $\int_0^6 f(x) dx$ is area of triangle^① + area rectangle^②

$$= \frac{1}{2} \times 2 \times 2 + 2 \times 2$$

$$= -6 \text{ (since below axis)}$$

iii) $P(x) = 2x^2 + 1$

(i, ii) satisfies

Q. 20. (i)

$$(i) \quad 2n = 11, \text{ etc.}$$

$$\begin{aligned} (ii) \quad S_{n-1} &= (1-1) + 4(n-1) \\ &= n^2 - 2n + 3 + 4n - 4 \\ &= n^2 + 2n - 3 \end{aligned}$$

$$\begin{aligned} b) \quad T_n &= S_n - S_{n-1} \\ &= n^2 + 4n - (n^2 + 2n - 3) \\ &= 2n + 3 \end{aligned}$$

$$\begin{aligned} c) \quad T_{10} &= 2 \times 10 + 3 \\ &= 23 \end{aligned}$$

(ii) 1st yr saves \$1000

2nd yr \$1200 etc

AP with $a = 1000$

$$d = 200$$

Want $S_n = 58000$.

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$58000 = \frac{n}{2} [2000 + (n-1) \times 200]$$

$$58000 = \frac{n}{2} \times 100 [20 + 2n - 2]$$

$$580 = n[9 + n]$$

$$\therefore n^2 + 9n - 580 = 0$$

$$(n + 29)(n - 20) = 0$$

$$n = -29 \text{ or } n = 20 \text{ (need } n \text{ pos.)}$$

Will take 20 years

(iii) GP $a = 32$

$$r = \frac{24}{32} = \frac{3}{4}$$

$$\begin{aligned} S_{\infty} &= \frac{a}{1-r} \\ &= \frac{32}{1-\frac{3}{4}} \end{aligned}$$

$$= 128$$

Teacher's Name

Student's Name (N°)

Question?

A) $f(x) = x^3 + 9x^2 + 20x + 3$

(i) $f'(x) = 3x^2 + 18x + 20$
 $= 3(x^2 + 6x + 8)$
 $= 3(x+2)(x+4)$

$f'(x) = 0$ when $x = -2$ or $x = -4$
 $y = -17$ $y = -13$

$f''(x) = 6x + 18$

$f''(-2) = 6(-2) + 18$
 $= 6 > 0 \Rightarrow \text{min}$

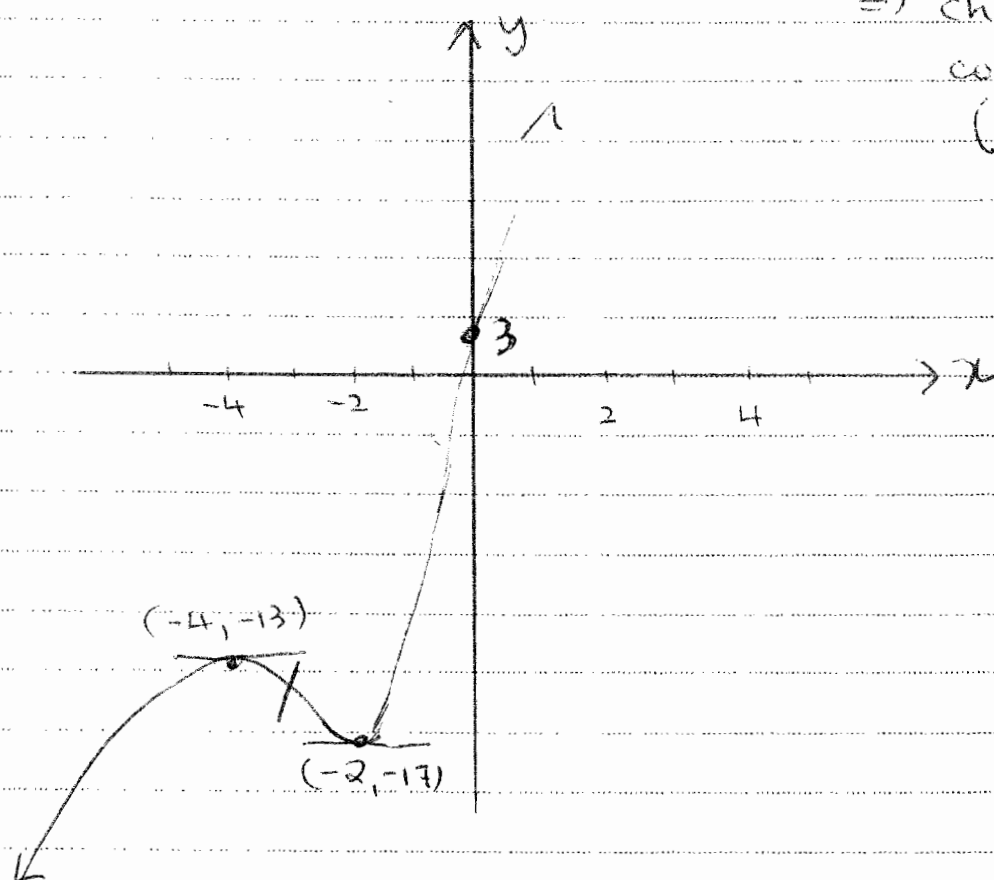
$f''(-4) = 6(-4) + 18$
 $= -6 \Rightarrow \text{max}$

y intercept is 3

(ii) $f''(x) = 0$ when
 $6x + 18 = 0$

ie $x = -3$

and changes sign
 $\Rightarrow$ change of
 concavity.
 $(-3, -15)$

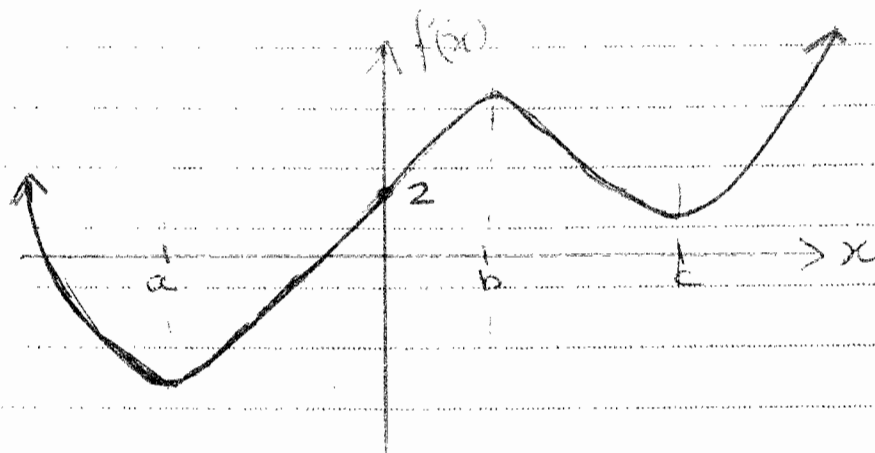


(iii) Decreasing for

Teacher's Name: _____

Student's Name/Nº: _____

- (15) (i) Stationary at $x = a$, if $f'(a) = 0$
(ii) Increasing when $f'(x) > 0$
 $a < x < b$, where $x > c$
(iii)



Question 4

A) (i) 2% per quarter

$$(ii) A_1 = 400,000(1.02)^1 - m$$

$$(iii) A_2 = [400,000(1.02) - m](1.02) - m \\ = 400,000(1.02)^2 - m(1.02 + 1)$$

End of first year will be 4 payments

∴ Want A_4

$$A_4 = 400,000(1.02)^4 - m(1.02^3 + 1.02^2 + 1.02 + 1) \\ = 400,000(1.02)^4 - m(1 + 1.02 + 1.02^2 + 1.02^3)$$

(iv) 12 yrs quarterly $\Rightarrow$ 48 payments
and $A_{48} = 0$

$$\therefore 0 = 400,000(1.02)^{48} - m(1 + 1.02 + \dots + 1.02^{47})$$

$$m = \frac{400,000(1.02)^{48}}{1 + 1.02 + \dots + 1.02^{47}} \leftarrow \text{GP with } a=1$$

$$r = 1.02$$

$$n = 48$$

$$S_{48} = \frac{1(1.02^{48} - 1)}{1.02 - 1}$$

$$\therefore m = \frac{400,000(1.02)^{48}}{(1.02^{48} - 1)} \times 0.02$$

$$= \$13041 \quad (\text{nearest dollar})$$

$$B) \quad i) \quad \begin{array}{c|c|c|c} x & 0 & a & 2a \\ \hline y & a+10 & a & a-10 \end{array}$$

$$ii) \quad 3200 \div \frac{a}{3} [(a+10) + (a-10) + 4a]$$

$$9600 \div a \times 6a$$

Question 5

$$A) \quad y = 2^x - 1 \quad \text{and} \quad y = 5x + 5$$

$$i) \quad \text{Test } x = -1 \quad y = (-1)^4 - 1$$

$$= 0$$

$$y = 5(-1) + 5$$

$$= 0$$

$$\text{Test } x = 2 \quad y = 2^4 - 1$$

$$= 15$$

$$y = 5(2) + 5$$

$$= 15$$

∴ Since $(-1, 0)$ and $(2, 15)$ satisfy each equation, these are the points of intersection.

$$ii) \quad \int_{-1}^2 (5x + 5) - (x^4 - 1) dx$$

$$= \int_{-1}^2 (-x^4 + 5x + 6) dx$$

$$= \left[-\frac{x^5}{5} + \frac{5x^2}{2} + 6x \right]_{-1}^2$$

$$= \left[-\frac{32}{5} + \frac{20}{2} + 12 \right] - \left[\frac{1}{5} + \frac{5}{2} - 6 \right]$$

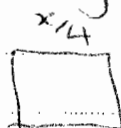
$$= -\frac{33}{5} + 26 - \frac{5}{2}$$

$$= 18 \frac{9}{10} \text{ u}^2$$

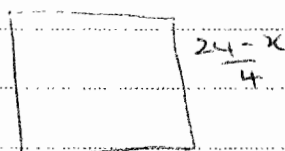
$$B) \quad i) \quad \begin{array}{|c|c|} \hline x & 24-x \\ \hline \end{array}$$

$$\text{length} = 24 - x$$

ii)



$$P = x$$



$$P = 24 - x$$

$$\text{Area} = \left(\frac{x}{4}\right)^2 + \left(\frac{24-x}{4}\right)^2$$

$$iii) \quad \frac{dA}{dx} = 2\left(\frac{x}{4}\right) \times \frac{1}{4} + 2\left(\frac{24-x}{4}\right) \times \left(-\frac{1}{4}\right)$$

Teacher's Name: _____

Student's Name: N° _____

$$\frac{dA}{dx} = \frac{x - 24 + 12x}{4}$$

$$= \frac{13x - 24}{4}$$

$$= \frac{x - 12}{4}$$

$$\frac{dA}{dx} = 0 \text{ when } \frac{x - 12}{4} = 0 \text{ i.e. } x = 12$$

$$\frac{d^2A}{dx^2} = \frac{1}{4} > 0 \Rightarrow \text{minimum}$$

∴ Minimum area when $x = 12$

Then

$$\begin{aligned} \text{Area} &= \left(\frac{12}{4}\right)^2 + \left(\frac{24 - 12}{4}\right)^2 \\ &= 9 + 9 \\ &= 18 \text{ u}^2 \end{aligned}$$